CHAPTER 1

Tang & Yang

Rong Tang

Yun Yang

2019–2025

Annals of Statistics JRSS

MCMC

1.

1.1.

.

D

$d \ll$
D

D

GAN
VAE

(1)

(2)

(3)

MCMC

Tang-
Yang

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Tang
&
Yang
2019–
2025

8

$MCMC$

1

2
VAE

3

MCMC

2.

2.1.

.

$\mathcal{M}^*$

$\mathbb{R}^D$

d

β-

$\beta \geq$
1

P^*

$\mathcal{M}^*$

$\mathrm{vol}_{\mathcal{M}^*}$

p^*

α-

$\alpha \in$
$[0, \beta -$
$1]$

p^*

$\mathbb{R}^D$

Lebesgue

Lebesgue

**Par-
ti-
tion
of
Unity**
Tang

$(U_\lambda, \varphi_\lambda)$

$\{\gamma_\lambda\}$

?]]tang2022minimax

?]]tang2023lowdim

Reach

reach

—

reach($\mathcal{M}$) =
inf{$r >$
0 :
$\mathcal{M}$

r-

}

2.2.

IPM.

KL

Hellinger

Tang

IPM

$$d_{\mathcal{F}}(\mu,\nu) = \sup_{f\in\mathcal{F}} \Big| \int_{\mathbb{R}^D} f(x)\,\mathrm{d}\mu - \int_{\mathbb{R}^D} f(x)\,\mathrm{d}\nu \Big|.$$

$\mathcal{F}$

γ-
Hölder

Sobolev

1-
Lipschitz

Wasserstein-
1

RKHS

MMD
γ

trade-
off

$\gamma =$
1

d_γ

Wasserstein-
1

$\gamma \to$
0^+

d_γ

?]]tang2025regression

2.3.

.
Tang

$$= n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}}.$$

$n^{-1/2}$

d

β-

d

Haus-
dorff

D

Lep-
ski

chain-
ing
peel-
ing
lo-
cal-
iza-
tion

Gir-
sanov

Poincaré

2.4.

.
Gauss-
ian

Langevin

$p_\sigma(x) = \int p_{\mathrm{data}}(y)\phi_\sigma(x - y)\,\mathrm{d}y$

L_∞

-

$\nabla \log p_t(x)$

t

$t =$
0

$t =$
T

?]]tang2024diffusion

$\mathcal{S}_{\mathrm{NN}}$

t

Girsanov

Gir-
sanov

SDE

3.

3.1. ?]]tang2022minimax

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Rong Tang and Yun Yang.

"Minimax Rate of Distribution Estimation on Unknown Submanifold under Adversarial Losses." ***Annals of Statistics*, 2022.**

D

GAN
VAE

γ

$$n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}},$$

α

β

d

D

partition of unity

VAE

**Tay-
lor**

**Lep-
ski**

Fano

**Varshamov-
Gilbert**

?]]tang2022minimax

?]]tang2023lowdim,
?]]tang2024diffusion

3.2. ?]]tang2023lowdim

—

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Rong Tang and Yun Yang.

"Estimating Distributions with Low-dimensional Structures Using Mixtures of Generative Models." ***ICML*****, 2023.**

VAE GAN

MLD-
MAE

K

-

ρ_k

1-
Wasserstein

α-

β-

d

$$O\Big(n^{-\frac{\alpha\wedge(\beta-1)+1}{2(\alpha\wedge(\beta-1))+d}}\Big) \vee O\Big(\frac{\log n}{\sqrt{n}}\Big),$$

$\alpha \leq$
$\beta -$
1

/

?]]tang2022minimax

?]]tang2022minimax

3.3. ?]]tang2023robust

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.

Rong Tang, Anirban Bhattacharya,

Debdeep Pati, and Yun Yang. “Robust Bayesian Inference on Riemannian Submanifold,” *JRSS B*, 2023.

Stiefel

Uncertainty Quantification

Bayesian RPE-TEL

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(1)

(2)

(3)

**Bernstein-
von
Mises
(BvM)**

**set-
ting**

**Me-
trop-
o-
lis
(RRWM)**

d

D

**con-
duc-
tance
pro-
file**

3.4. ?]]tang2023twosample
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**Rong
Tang
and
Yun
Yang.
“Min-
i-
max**

Nonparametric Two-Sample Test under Adversarial Losses,” *AISTATS*, 2023.

MMD

$O(n^2)$

Besov

Besov

1

ℓ_1
ℓ_2

Wasser-
stein

1

2

$O(n \log n + n^{2d/(4\alpha+d)})$

MMD

$O(n^2)$

3.5. ?]]tang2024diffusion

—

.

Rong Tang and Yun Yang. "Adaptivity of Diffusion Models to Manifold Structures," ***AISTATS*****, 2024.**

SOTA

Langevin

–

d

D

Langevin

**Gauss-
ian**

$\tilde{O}(\sigma)$

σ

n

d

L_∞

$-$

1-
Wasserstein

$$\frac{1}{\sqrt{n}} \vee n^{-\frac{\alpha+1}{2\alpha+d}}.$$

?]]tang2022minimax

?]]tang2022minimax

3.6. ?]]tang2024mala — Metropolis-Adjusted Langevin

.

Rong Tang and Yun Yang. "Metropolis-Adjusted Langevin Algorithm: Optimal Dimension Dependency and Beyond."

***ICML*, 2024.**

MALA

MCMC

$O(d^{1/3})$

s**-**
conductance
pro-
file

MALA

$O(d^{1/3})$

**burn-
in**

s-conductance profile

conductance

Chen et al. (2020)

conductance profile

**warm-
start**

$\log M_0$

$\log \log M_0$

?]]tang2023robust

?]]tang2023robust

BvM

BvM

MALA

3.7. ?]]tang2025regression

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Rong Tang and Yun Yang. "Minimax Optimal Rates for Regression on Manifolds and Distributions," arXiv, 2025.

X

Y

$\mu^*_{Y|X}$

X

Y

IPM

(1)
Regime
1

Lebesgue

$$n^{-\frac{\alpha_X}{2\alpha_X+d_X}}+n^{-\frac{\alpha_Y+\gamma}{2\alpha_Y+D_Y+\frac{\alpha_Y}{\alpha_X}\cdot d_X}}.$$

(2)

Regime 2

:

$\mathcal{M}_{Y|X}$

X

$n^{-\gamma\beta_Y/d_Y}$

(3)

Regime 3

:

$\mathcal{M}_{Y|x}$

x

$n^{-\gamma/(\frac{d_Y}{\beta_Y}+\frac{d_X}{\beta_X})}$

γ-**Hölder**

IPM
d_γ

$\gamma =$
0

1-
Wasserstein

$\gamma =$
1

3.8. ?]]tang2025conditional
—

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Rong
Tang
and
Yun
Yang.

"Conditional Diffusion Models are Minimax-Optimal and Manifold-Adaptive," arXiv, 2025.

?]]tang2024diffusion

(1)

$\alpha_X \in$
$[0, 1]$

(2)

X

Y

**Wasser-
stein**

(d_X, d_Y)

(D_X, D_Y)

**lo-
cal-
iza-
tion
and
peel-
ing
tech-
niques**

**Gir-
sanov**

?]]tang2024diffusion,
?]]tang2025regression

?]]tang2024diffusion

$\rightarrow$
?]]tang2025regression

$\rightarrow$

4.

**Tang-
Yang**

**2019–
2025**

4.1.